

Keith Walsh

Analytics Engineer

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PROFILE

Engineer with 15 years across data, e-commerce and enterprise systems. At Portwest I built the data layer from scratch: a tested, 104-model dbt project over a global Infor M3 ERP estate, and the MCP server that lets AI tools read the ERP schema safely. Before that, four years of SQL and Python on Snowflake at Pitney Bowes, recalculating client invoices across pricing structures and building client-facing Tableau and Power BI dashboards. Equally comfortable tracing a data defect across four systems or translating a Finance requirement into a tested, documented data product.

SKILLS

Analytics Engineering	dbt, layered modelling (staging → intermediate → marts), generic and custom tests, ELT/ETL, incremental loads, data contracts, data validation
Languages	SQL, Python, PHP, JavaScript/TypeScript, HTML/CSS
Databases & Cloud	Snowflake, Azure SQL Managed Instance, SQL Server, MySQL, Oracle
BI & Dashboards	Tableau, Power BI, reporting views, dashboard design
Engineering Practice	Git/Bitbucket, pull-request review, command line, Pest/PHPUnit, regression testing, job scheduling and observability, Agile
AI & Agent Tooling	Model Context Protocol (MCP) server development, LLM-facing API design, agent skill authoring, retrieval quality tuning, Microsoft Copilot integration
Applications & Systems	Laravel, REST API design, secure code review, Infor M3 (ION API, MDBREADMI/EXPORTMI), Akeneo PIM, EDI, Salesforce, Jira

EXPERIENCE

Digital Support Engineer

Portwest, Westport, Co. Mayo | Apr. 2025 – Present

- Data modelling & architecture:** Designed and maintain a 104-model dbt project over M3 ERP data (57 staging, 16 intermediate, 15 mart, 16 admin) with generic and custom tests, plus the SQL views, Azure SQL Managed Instance structures and Python automation behind daily refreshes, incremental loads and large-scale exports.
- Single source of truth & documentation:** Authored the reference documentation for 149 M3 tables, drawing an explicit line between vendor-defined field semantics and measured local usage so downstream reports and models do not misread a field.
- Data layer for AI tools:** Built a Model Context Protocol server in Python giving LLMs structured, read-only access to the M3 schema: six tools over 141 tables and 2,936 field definitions, shipped with a self-test harness. Authored six reusable agent skills and the API endpoints exposing M3 data to Copilot.
- Data quality & cross-system analysis:** Resolve defects spanning the ERP, webshop, PIM and warehouse systems, including the quote-to-promotional-price pipeline, EDI order failures and multi-warehouse stock across US and EU sites. Built a cron-health dashboard after diagnosing that scheduled jobs were failing silently.
- Application engineering & security:** Principal developer on the internal Laravel platform (217 classes, ~46% of commits, branch-and-pull-request workflow); found and fixed an unauthenticated SQL injection on a public endpoint, audited access control across all report endpoints and added regression tests to an 88-file Pest suite.

Data Analyst

Pitney Bowes, Dublin | Mar. 2021 – Dec. 2024 | *role ended with the closure of the Global E-commerce division*

- Data-driven invoicing:** Built a rerate tool in Python and SQL over Snowflake to recalculate client invoices across multiple pricing structures and data sources.

- **Process automation:** Developed automated Python/SQL scripts to detect invoice errors and validate refund calculations, cutting manual effort by over 50%.
- **Client-facing dashboards:** Designed and maintained internal and client-facing Tableau and Power BI dashboards, including dashboards supporting client retention strategy and analysis that accelerated claim adjudication.
- **Loss prevention:** Developed algorithms and reporting tools to streamline the Loss Prevention team's operations and inform risk assessment.

Client Support Manager

Pitney Bowes, Dublin | Nov. 2019 – Feb. 2021

- Led a team of 13 agents, owning performance reviews, development plans and resource allocation.
- Built automated KPI reporting to monitor performance and surface service delivery gaps.
- Refined cross-functional workflows with department leaders, reducing escalations by 25%.

Client Support Business Analyst

Pitney Bowes, Dublin | Jun. 2018 – Oct. 2019

- Developed a browser extension automating agent tasks, improving productivity by 23% and increasing accuracy.
- Created the Agent Proficiency Score (APS) system, enabling performance tracking and weekly trend analysis.
- Deployed custom scripts that reduced average handling times by 15%.

Payments & Risk Management Analyst

BetBright / 888, Dublin | Apr. 2015 – Jun. 2018

- Led risk assessment and fraud prevention initiatives, reducing payment disputes by 28%.
- Built and managed a team of three Fraud & Payments Analysts, introducing structured training and KPI tracking.
- Streamlined KYC/AML processes, halving verification times while maintaining compliance.

Earlier Roles

- **Fraud Analyst**, GVC / Sportingbet, Dublin | Jan. 2013 – Mar. 2015 — analysed transaction data in Oracle to identify fraud patterns and prevent losses.
- **Payments Administrator**, GVC / Sportingbet, Dublin | Nov. 2011 – Jan. 2013
- **Billing Administrator**, Airtricity, Dublin | Oct. 2010 – Oct. 2011

EDUCATION

- **B.Sc. Engineering with Management**, Trinity College Dublin, 2010
- **Google Data Analytics Professional Certificate**, Google, 2025
- **GitHub Certification**, GitHub, 2025
- **Certificate in Business Accounting**, Chartered Institute of Management Accountants, 2013
- **Certificate in Financial Crime Prevention**, International Compliance Association, 2016
- **Specialist Certificate in Money Laundering Risk in Betting**, International Compliance Association, 2015

AWARDS

- **Pitney Bowes Recognition Award (2023–2024)**, multiple nominations for cross-functional collaboration and support; **Innovation Award 2021**, 3rd place, Continuous Innovation

COMMUNITY

- **GitHub:** maintain public repositories and publish packages to npm and PyPI — github.com/keithwalsh
- **Stack Overflow:** contribute answers on SQL and Python
- **Personal:** Home Assistant automation; time with my daughter and the outdoors